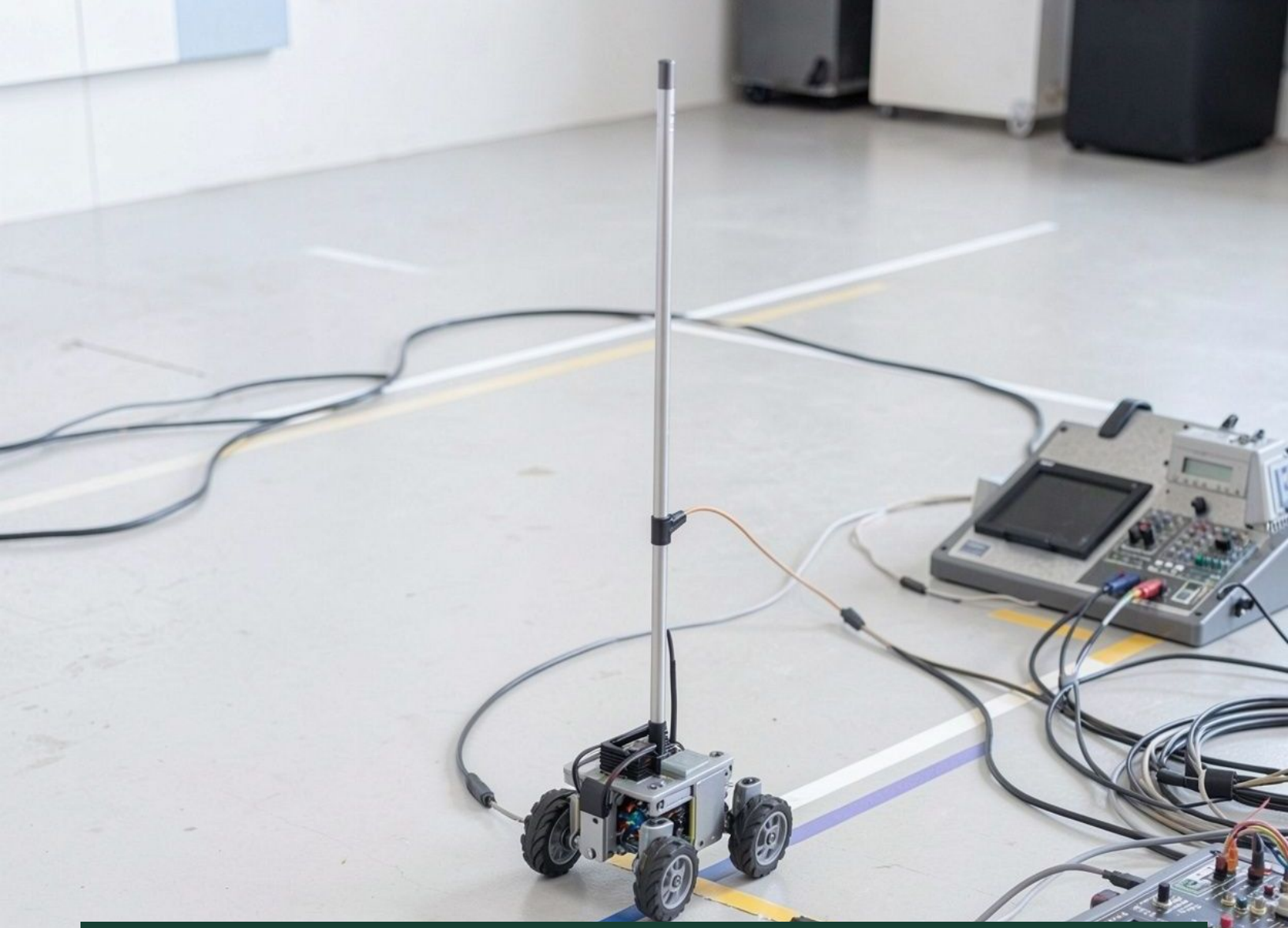


EDGE AI AND ROBOTICS · DEEP REINFORCEMENT LEARNING

Enhanced DQN for Stable and Sample-Efficient Control

Empirical evaluation and mathematical remediation of overestimation bias, sample inefficiency, and policy instability in Deep Q-Networks.

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EXECUTIVE SUMMARY

Enhanced DQN improves both control quality and learning efficiency

Three targeted interventions address three structural weaknesses in vanilla DQN: Double Q-Learning reduces overestimation, Prioritized Experience Replay concentrates updates on informative transitions, and soft target updates smooth the learning signal, supported by Huber loss.

439.45	−78.55	4,529
CartPole-v1 final 20-episode average, versus 94.20 for vanilla DQN	Acrobot-v1 final 20-episode average, versus −104.05	fewer interaction steps on Acrobot-v1 across training

The reported CartPole improvement is 366.5%. The enhanced method takes longer to train, but produces more stable control and more efficient task progress.

Vanilla DQN is powerful but structurally vulnerable

Introduced by Mnih et al. in 2013, DQN made deep value learning practical through two stabilizing pillars: a replay buffer sampled uniformly and a periodically copied target network.

BELLMAN OPTIMALITY

$$Q^*(s,a) = \mathbb{E}\left[r + \gamma \max_{a'} Q^*(s',a') \mid s,a\right]$$

The maximization operator is central to action selection, but it also amplifies noisy value estimates.

BASELINE CONFIGURATION

N = 50,000

circular replay buffer

C = 200

steps between hard target updates

Three bottlenecks explain instability and wasted experience

Overestimation bias

The same target weights select and evaluate the greedy action, so noisy estimates are amplified by the maximization operator.

Uniform replay

Every transition receives probability 1/|D|, so mastered zero-error states compete equally with informative milestone states.

Hard target updates

Periodic parameter copies create discontinuities in the target and can produce loss oscillations and erratic gradients.

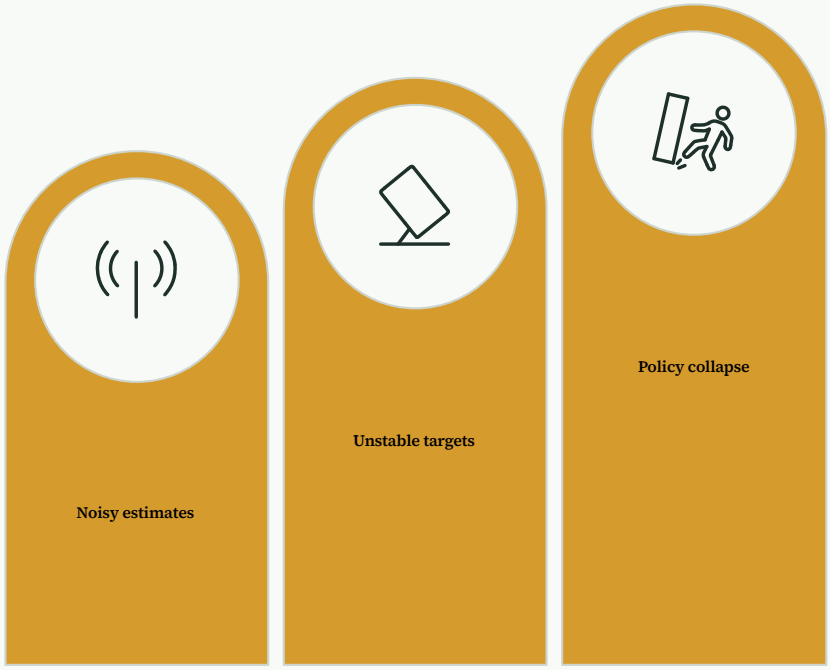


Figure 1 · Causal path from estimation noise to unstable control.

Enhanced DQN decouples selection, prioritizes learning, and smooths targets

The enhanced design changes the source of the target, the order of replay, and the way target parameters move.

Double Q-Learning

The online network θ selects argmax a′ Q(s′, a′; θ); the target network θ^- evaluates that action.

Prioritized replay

Sample priority is (|δi| + ε)^α with α = 0.6; importance-sampling correction anneals β from 0.4 to 1.0.

Soft targets

After each gradient step, $\theta^- \leftarrow \tau\theta + (1 - \tau)\theta^-$ with τ = 0.005.

Huber loss

The penalty becomes linear for |δ| > 1.0, limiting gradient surges on surprising transitions.

Figure 2 · Four interventions, each mapped to one instability or efficiency bottleneck.

The implementation turns each remediation into an explicit algorithmic step

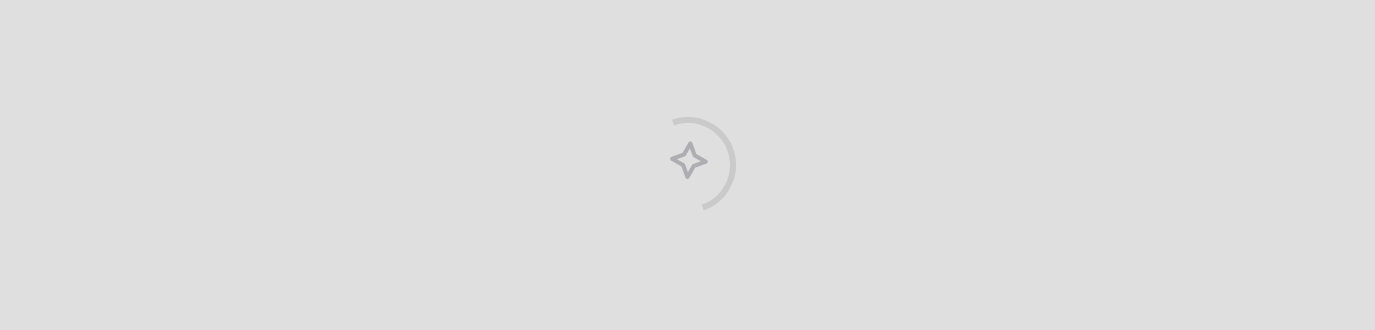


Figure 3 · The training loop makes every remediation observable.

Efficient priority access

The custom binary _SumTree supports priority queries and updates in O(log N), rather than scanning an entire buffer in O(N).

DOUBLE DQN TARGET

$$Y_{\text{DoubleDQN}} = r_t + \gamma(1 - d_t)Q(s_{t+1}, \arg\max_{a'} Q(s_{t+1}, a'; \theta); \theta^-)$$

Evaluation compares identical training budgets across two control tasks

Both algorithms were trained for 300 complete episodes from identical random seeds. CartPole-v1 tests balance; Acrobot-v1 tests swing-up control.

TASK	ALGORITHM	FINAL 50-EP AVG	FINAL 20-EP AVG	PEAK REWARD	TOTAL STEPS	TRAINING TIME
CartPole-v1	Vanilla DQN	166.50	94.20	500.0	28064	61.9s
CartPole-v1	Enhanced DQN	390.90	439.45	500.0	44614	151.7s
Acrobot-v1	Vanilla DQN	-94.32	-104.05	-62.0	49232	137.3s
Acrobot-v1	Enhanced DQN	-84.22	-78.55	-62.0	44703	164.7s

Table 1 · Experimental comparison metrics. Final averages, peak reward, total interaction steps, and training time are reported for both tasks.

CartPole stability and Acrobot efficiency are the clearest gains

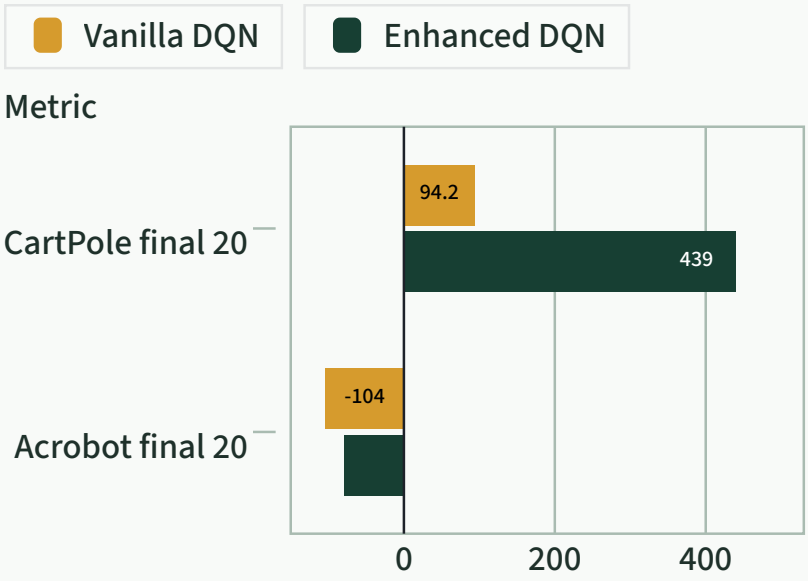


Figure 4 · Final 20-episode average reward across the two tasks.

Read the result

CartPole-v1. Vanilla DQN collapses after episode 180. Enhanced DQN maintains a near-optimal final 20-episode average of 439.45 against the 500-point ceiling.

Acrobot-v1. Enhanced DQN reaches -78.55 and uses 4,529 fewer interaction steps.

Training time is the cost of the enhanced method: 151.7s versus 61.9s on CartPole-v1, and 164.7s versus 137.3s on Acrobot-v1.

Viva questions connect the implementation to the theory

01

Why Double DQN?

Selection and evaluation use separate networks, reducing positive maximization bias.

02

Why importance sampling?

Non-uniform replay changes the state distribution, so IS weights correct the resulting gradient bias.

03

Why a SumTree?

O(log N) priority sampling and updates make PER tractable for a buffer of 50,000 transitions.

Answer by linking each implementation choice to the failure mode it remediates: bias, wasted experience, or unstable targets.

CONCLUSION

Stability and sample efficiency improve together

The project diagnoses the structural weaknesses of vanilla DQN and shows how an Enhanced DQN combines decoupled value estimation, informative replay, smooth target tracking, and robust loss shaping.

Future extensions

Dueling Networks, multi-step returns, and Distributional RL such as QR-DQN.

Practical trade-off

The Enhanced DQN takes longer to train, but produces more stable control and more efficient task progress.